

1 . What do you understand by Natural Language Processing?

Natural Language Processing (NLP) is a branch of **Artificial Intelligence (AI)** that enables computers to understand, interpret, process, and generate human language in both text and speech form. It combines techniques from computer science, linguistics, and machine learning to help machines communicate with people naturally.

NLP is used in many everyday applications such as chatbots, language translation, voice assistants, sentiment analysis, spam detection, and text summarization. Its main goal is to bridge the gap between human communication and computer understanding.

Examples: Voice assistants like Siri and Google Assistant, language translators, email spam filters, and customer support chatbots.

2. What are the steps involved in solving an NLP problem?

The main steps involved in solving an **NLP (Natural Language Processing)** problem are:

1. **Data Collection**

Gather text or speech data from sources such as documents, websites, social media, or customer reviews.

2. **Text Preprocessing**

Clean and prepare the text by:

- Converting text to lowercase
- Removing punctuation and special characters
- Tokenization (splitting text into words or sentences)
- Removing stop words
- Stemming or lemmatization

3. **Feature Extraction**

Convert text into numerical form using techniques such as:

- Bag of Words (BoW)
- TF-IDF
- Word Embeddings (Word2Vec, GloVe)

4. **Model Building**

Choose and train an appropriate NLP or machine learning model based on the task, such as text classification, sentiment analysis, or translation.

5. **Model Evaluation**

Test the model's performance using metrics like accuracy, precision, recall, and F1-score.

6. **Deployment and Monitoring**

Deploy the trained model for real-world use and monitor its performance, updating it as needed.

3. What is an ensemble method in NLP? With Example.

An **ensemble method in NLP** is a technique that combines the predictions of **multiple machine learning or NLP models** to improve overall accuracy and performance. Instead of relying on a single model, an ensemble uses the strengths of several models to produce better and more reliable results.

Common ensemble techniques include:

- **Bagging** (e.g., Random Forest)
- **Boosting** (e.g., AdaBoost, Gradient Boosting)
- **Stacking** (combining different models)

Example:

In a **sentiment analysis** task, three different models—such as **Naive Bayes**, **Support Vector Machine (SVM)**, and **Decision Tree**—can be trained on the same dataset. Their predictions are then combined using majority voting. If two out of three models predict that a movie review is positive, the ensemble model classifies it as positive. This often gives better accuracy than using a single model.